Daniel Hardesty Lewis

Staff ML Engineer – production training, inference, evaluation, reliability

Specializing in automated validation frameworks and scalable distributed infrastructure. Delivered consumer pricing and statewide resiliency platforms across geospatial & time-series workloads.

Core Expertise

8 years	**ML Engineering**: Build and operate AI/ML training and inference pipelines, using rigorous validation protocols to ensure model generalization and reliability
8 years	**Backend & Data**: Build schemas and ETL; ship API-backed services using Python, SQL, and Postgres/PostGIS
8 years	**Distributed Systems**: Scale batch workloads across distributed systems, profile bottlenecks, and optimize latency & throughput with explicit cost-performance tradeoffs
10 years	**Geospatial AI**: Spatiotemporal modeling for risk, planning, and hazard decision-support use cases
8 years	**Technical Leadership**: Drive cross-functional execution via specifications and governance; mentor engineers through design reviews, code reviews, and instruction

Technologies

Modeling	PyTorch; TensorFlow; scikit-learn; Transformers; LoRA/QLoRA/PEFT
LLM apps	embeddings; RAG; LangChain; LangGraph; DSPy; FAISS
Data stack	Python; SQL; Pandas; Polars; PyArrow; Parquet; Postgres; PostGIS; Redis; Elasticsearch; Supabase
Infrastructure	Linux/Unix; Bash; Git; Docker; Kubernetes; Modal; GCS; Ray; Spark; Dask; Airflow; CI/CD; observability
Additional	CUDA; C++; TypeScript; React

Experience

Professional

Apr. 2025 – Present
Founder, *Homecastr*, New York, NY
Consumer AI forecasting platform serving thousands of monthly active users. Owned the end-to-end ML architecture from data ingestion to web delivery.

- Architected a nationwide neighborhood-level diffusion forecasting model, **matching Zillow's 7.3% 0-year error benchmark at a 1-year forecast horizon** and maintaining a **35.6% median error over 4 years**.
- Engineered architecture incorporating **spatial inducing tokens**, **per-horizon loss normalization**, and **DDIM sampling** to balance predictive precision with calibrated price trajectories.
- Developed predictive validation optimizing PIT uniformity, interval coverage, and sharpness to **rigorously validate forecast reliability** before production delivery.
- Owned the end-to-end pipeline from data ingestion to real-time serving and refined product strategy through direct feedback from **a16z and MetaProp**.

Oct. 2024 – Sep. 2025
Technical Advisor, *PoliBOM*, Seattle, WA
AI tariff mitigation and trade compliance product for manufacturers, centered on agentic BOM analysis and policy scenario simulation.

- Engineered an agentic retrieval-augmented pipeline for BOM parsing and tariff simulation, replacing manual trade compliance workflows with predictable multi-step reasoning.
- Directed development of a **conversational AI interface** enabling decision-makers to instantly **retrieve BOMs and receive proactive compliance alerts**.
- Executing customer discovery with ScoutyAI and Columbia economists, and secured early market validation by presenting the prototype to KPMG.

Nov. 2023 – Mar. 2025
Founder, *Summit Geospatial*, New York, NY
High-resolution elevation data and web delivery for engineering and hazard workflows.

- Engineered a **statewide seamless Texas DEM mosaic**, resampling 0.5 m LiDAR sources to 1.2 m via nearest-neighbor across **70+ state, federal, and local elevation datasets**.
- Developed the web distribution platform end-to-end including **data pipeline, tiling, hosting, delivery UX** to support engineering and planning use cases.

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Aug. 2021 – Aug. 2023 **Senior Data Scientist & Technical Lead**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- Spun out Summit Geospatial alongside UT's VPs of IP to commercialize state-level terrain models from the $40M Texas Disaster Information System (TDIS), delivering precision decision-support for civil engineering.
- Developed inter-agency RFP and MOU frameworks with state and federal partners, translating technical requirements into scope and milestones facilitating Real-Time Inundation Mapping and TDIS.
- Scaled climate and flood models on supercomputers, executing 800,000-node distributed jobs while managing $1M+ compute budgets and federal partnerships with NOAA, NSF, USACE, GLO.
- Developed efficient methods to produce high-resolution 1m flood maps from National Water Model outputs for Texas Emergency Management's Real-Time Inundation Mapping.
- Partnered with the US Army Corps of Engineers to statistically model compound flood hazards in coastal Texas, integrating high-resolution topographic data with hydrological models.
- Contributed fundamental research supporting Paola Passalacqua's 2022 EGU Bagnold Medal, one of geomorphology's highest honors.
- Served as Technical Reviewer for Frontera Doctoral Fellows (2023) and Large-Scale Computing Pathways (2022) applications, evaluating $10M+ in computing resource allocations on one of the world's largest supercomputers.

Feb. 2018 – Jul. 2021 **Data Scientist & Research Engineer**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- Competed in DARPA World Modelers to improve disaster resiliency and food security modeling in East Africa.
- Mentored Petrobras engineers in deep learning and HPC to enable institutional technology transfer.
- Received research recognition from AAAI President Yolanda Gil during her farewell address for contributions to automated scientific modeling.

Research

May 2025 – Aug. 2025 **Research assistant to Dir. Electrical Engineering Zoran Kostic**, *Columbia University*, New York, NY

- Math benchmarking and finetuning of Deepseek's GRPO algorithm

Jan. 2025 – Oct. 2025 **Research assistant to Dir. Financial Engineering Ali Hirsa (Industry-sponsored research)**, *Columbia University*, New York, NY

- Developed a multi-asset CVAE latent factor model with Skew-T Mixture priors; achieved 85% R^2 vs. commercial SaaS (75%) and Fama-French (58%) on backtested holdout.
- Derived a tractable, axiom-compliant per-feature SHAP algorithm using Hessian-based recursive clustering to provide stable temporal regime-dependent attributions for deep learning models.

July 2019 – Aug. 2019 **Summer research intern to Pres. AAAI Yolanda Gil**, *University of Southern California*, Los Angeles, CA

- Integrated hydrological models into the MINT Platform during a year-long competition with MITRE to provide DARPA the "Airflow for science"

Teaching

Jan. 2018 – Dec. 2020 **Co-instructor**, *The University of Texas at Austin*, Austin, TX

Helped design and teach graduate-level courses: *Machine Learning for the Geosciences* for Petrobras engineers, *Intelligent Systems for the Geosciences*, and *Scientific Computation*.

Select Publications

2021 Gil, Y., et al. (incl. **D. Hardesty Lewis**), "Artificial Intelligence for Modeling Complex Systems: Taming the Complexity of Expert Models to Improve Decision Making". *ACM TiiS*, 2021. DOI: 10.1145/3453172

2019 Garijo, D., et al. (incl. **D. Hardesty Lewis**), "An intelligent interface for integrating climate, hydrology, agriculture, and socioeconomic models". *ACM IUI '19*, 2019. DOI: 10.1145/3308557.3308711

Education

expected 2026 **M.S., Urban Planning**, *Columbia University*, New York City, NY

2017 **B.S., Pure Mathematics**, *University of Texas*, Austin, TX

Key Projects

oppsAlert Developed the first property- and owner-level ML model of NIMBYism, using 15+ years of 15,000+ points of individual homeowner protest data provided by the City of Austin.

M&A predictor Built M&A prediction model (Oct.–Dec. 2025) with Prof. Eric Talley on 1.5M Compustat firm-quarters; long-only portfolio beat S&P 500 by 2%/yr (2004–2020), peaking at 6%/yr.

Activities

Sailing Volunteer Atlantic SAR of the Cheeki Rafiki

Bikepacking US West & East Coasts

Languages

Spanish Native Proficiency

French Professional Working Proficiency

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